

FUNCTIONAL REQUIREMENT SPECIFICATION OF NIDDO QUICK APP (React Native Android & iOS App)



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1. Project Overview

Niddo Quick is an AI-enabled roadside assistance and auto spare parts marketplace platform developed for Niddo Roadlines Pvt. Ltd.

The system connects:

- Drivers experiencing vehicle breakdowns
- Verified mechanics
- Spare part vendors
- Delivery partners
- Centralized admin control

The platform enables real-time breakdown assistance, smart spare part procurement, automated partner assignment, commission-based revenue management, and AI-powered decision-making.

The solution consists of:

- Customer (Driver) Mobile App(Android+ios)
- Mechanic Mobile App(Android+ios)
- Delivery Partner Mobile App(Android+ios)
- Vendor Web Panel
- Admin Web Panel

2. Project Objective

The primary objectives of the Niddo Quick platform are:

- Provide instant roadside assistance through AI-based mechanic assignment.
- Create a centralized virtual spare parts store routing engine.
- Enable automated vendor and delivery partner assignment within defined SLA.
- Implement a commission-based monetization and settlement system.
- Ensure scalability to support multi-city operations and 5,000+ concurrent users.
- Provide full transparency through tracking, reporting, and audit logs.
- Ensure high system availability ($\geq 99\%$ uptime).
- Maintain full data ownership and security compliance.

3. Users (Actors)

Actor	Description	Key Responsibility
User (Driver / Customer)	Individual vehicle owner or driver who uses the Niddo Quick mobile application to request roadside assistance and purchase spare parts.	• Register & manage profile and vehicles • Raise breakdown requests • Track mechanic in real-time • Purchase spare parts • Track spare part delivery • Make payments (Online/COD) • Provide ratings & reviews • Raise disputes if required
Mechanic	Verified service professionals registered on the platform to provide roadside assistance and vehicle repair services.	• Register and upload required documents • Set availability status (Online/Offline) • Accept or reject service requests • Navigate to breakdown location • Perform inspection and repair • Generate service bill • Update job status • View earnings and payout details
Vendor (Spare Part Store Partner)	Registered spare parts seller integrated into Niddo Virtual Store system to supply auto parts for breakdown services and direct orders.	• Register store and provide GST/bank details • Manage inventory (Add/Update/Delete parts) • Accept or reject spare part orders • Pack and prepare order for pickup

		• Monitor stock levels • View sales and commission reports • Receive AI-based inventory alerts
Delivery Partner	Registered logistics partner responsible for picking up spare parts from vendors and delivering to customer locations.	• Register and upload vehicle & ID details • Accept delivery assignments • Pick up order from vendor • Deliver order to customer • Collect COD (if applicable) • Upload proof of delivery • View earnings and payout reports
Admin	System administrator managing the complete Niddo Quick ecosystem including users, partners, AI engine, commission structure, and analytics.	• Approve/reject mechanics, vendors & delivery partners • Monitor breakdown requests & spare part orders • Override AI decisions (manual assignment) • Manage disputes and refunds • Generate revenue and performance reports • Monitor system KPIs & AI logs

4. Functional Requirements

4.1 .(User) Functional Requirements

4.1.1 User Registration :

Users can create an account using their mobile number.

Functional Flow:

1. The user enters a mobile number.
2. The system sends OTP.
3. The user verifies OTP.
4. User enters:
 - Full Name
 - Email (Optional)
 - Vehicle Details (Optional at signup)
5. An account is created.

Validation:

- Mobile number must be unique.
- OTP must expire in 2–5 minutes.
- Retry OTP limit: 3 times.

4.1.2 Login :

- Login via mobile number + OTP
- Auto-login if session active
- Logout option available



4.1.3 Dashboard (Home Screen):

User dashboard will show:

- Book Mechanic
- Order Spare Parts
- Emergency Assistance
- My Bookings
- My Orders
- Notifications
- Spare part Booking

4.1.4 Vehicle Management :

a. Add Vehicle:

- Vehicle Type (Car/Auto/Truck, Etc)
- Brand
- Model
- Registration Number
- Fuel Type

Note:

- Multiple vehicles allowed.
- One vehicle marked as default.

4.1.5 Mechanic Booking Module :

a. Search Mechanic

Filters:

- Location (GPS auto-detect)/ Manual
- Service type (Engine, Tire, Battery etc.)
- Rating
- Availability

System shows:

- Mechanic name
- Rating
- Distance(near by)
- Estimated arrival time
- Service charge (if predefined)

Note- Initially system will search mechanic, vendor within 15 km; if service is not available, then the system automatically increases the radius next 10 km during each search

b. Book Mechanic

Booking Flow:

- Select service.
- Select vehicle.
- Choose time:
 - Instant
 - Schedule later
- Confirm location.



- View estimated cost.
- Confirm booking.

c. Booking Status Flow

Booking Status:

- Requested
- Accepted
- Mechanic On the Way
- Arrived
- Work in Progress
- Completed
- Cancelled

d. Live Tracking

- Track mechanic via GPS
- View ETA updates
- Call mechanic option (In App Call)
- In-app chat

e. Cancel Booking

- Free cancellation before the mechanic starts
- Cancellation charge after mechanic's arrival or in the way the user need to give the cancellation charges.

4.1.6 Spare Parts Ordering:

a. Browse Parts

User can:

- Search parts
- Filter by vehicle model
- View product details
- Check stock availability

Product Info:

- Image
- Price
- Stock
- Vendor name
- Delivery time

b. Add to Cart

- Add multiple products
- Update quantity
- Remove item

c. Checkout Flow

- Confirm address (Address confirmation on current location)
- Select payment method
- Apply coupon (if available)

- Confirm order

d. Order Status Flow

Order Status:

- Order Placed
- Confirmed
- Packed
- Out for Delivery
- Delivered
- Cancelled

User can:

- Track delivery partner
- Call delivery partner

4.1.7 Payment Module:

a. Payments Flow

- Show final bill
- User selects payment method
- Redirect to payment gateway
- Payment success/failure screen
- Generate invoice

Note- Only Online payment will be available for the spare parts booking.

4.1.8 Notifications Module:

a. User receives:

- Booking updates
- Order updates
- Offers
- Payment confirmation

b. Delivery channels:

- Push notification
- SMS
- In-app notifications

4.1.9 Rating & Review

After service/order completion:

User can:

- Rate (1–5 stars)
- Write review
- Report issue

4.1.10. My Bookings Section

User can view:

- Active bookings
- Past bookings



- Cancelled bookings
- Download invoice

4.1.11. My Orders Section

- Active orders
- Order history
- Invoice download
- Reorder option

4.1.12. Profile Management

User can:

- Edit name
- Change phone (OTP verification)
- Manage vehicles
- Manage addresses
- Delete account

4.1.13. Address Management

User can:

- Add multiple addresses
- Use GPS location
- Mark default address

4.1.14. Logout

4.2 .(Mechanic) Functional Requirements

4.2.1 Registration & Onboarding

4.2.1.1 . Registration Fields :

- Full Name (Mandatory)
- Mobile Number (OTP verification)
- Email (Optional)
- Profile Photo
- Aadhaar Number
- Document (Like an Aadhar card, Training Certification)
- Experience (Years)
- Service Types (Car/Bike/Truck/etc.)
- Service Radius (Default 25 KM, Max 100 KM)
- Bank Details (For settlement)
- Garage Address (If any)

4.2.1.2. Approval Flow :

- Mechanic submits registration
- Status = Pending
- Admin reviews documents
- Admin can:
 - Approve
 - Reject (with reason)
- Only approved mechanics can go live

Status Options:

- Pending



- Approved
- Rejected
- Suspended

4.2.1.3. Dashboard (Home Screen):

Features:

- Go Online / Offline Toggle
- Today's Bookings
- Active Job
- Earnings Summary
- Rating Overview
- Notification Center

4.2.1.4. Booking Management :

a. Booking Flow:

Booking notification received

View:

- User name
- Vehicle details
- Location
- Distance
- Estimated payout

Accept / Reject (within defined time limit)

If accepted → Status: Assigned

Note - Only online payment will be applicable for the spare parts. And the COD and online will be applicable for the mechanic booking.

b. Job Status Flow :

Status	Action By Mechanic
Assigned	Accepted booking
Enroute	Start navigation
Reached	Arrived at location
Work Started	Begin repair
Parts Required	If spare part needed
Work Completed	Repair finished
Payment Pending	If COD
Completed	Payment confirmed

4.2.1.5. Payment Handling:

a. Online Payment

- Auto marked as Paid after gateway success

b. COD Payment

Flow:

1. Mechanic collects cash
2. Clicks "Cash Received"
3. Status → Completed

4.2.1.6. Commission Logic:

Mechanic Service Commission:

- 12% OR ₹150 (Whichever lower)
- Configurable from admin panel

Deduction:

- Auto deducted before settlement

Mechanic Earnings Screen Shows:

- Gross Amount
- Commission
- Net Payout
- Settlement Status

Note - The commission Criteria will be applied only by the admin side.

4.2.1.7. In-App Communication:

- In-App Chat with User
- In-App Call
- No direct phone number visibility
- Chat history stored

4.2.1.8. Ratings & Reviews:

After job completion:

- User rates mechanic (1–5 stars)
- Feedback text
- Rating impacts AI ranking system

Mechanic can:

- View average rating
- View individual feedback

4.2.1.8. Mechanic Spare Parts Ordering Module:

a. Purpose

Mechanic can:

- Order spare parts required for ongoing job
- Order parts for personal stock
- Compare vendor prices
- Get delivery at job location or garage

b. Ordering Flow

- Mechanic clicks “Order Parts”
- Search product
- Filter by:

- Vehicle type
- Compatibility
- Price
- Vendor
- Rating
- View product details:
 - Price
 - Vendor name
 - Delivery time
 - Stock
- Add to cart
- Select delivery location:
 - Current job location
 - Garage address
- Payment mode:
 - Online
 - COD
- Place Order
- Delivery partner auto assigned

c. Order Status Flow:

Status	Description
Placed	Order created
Accepted	Vendor confirmed
Packed	Ready
Out for Delivery	Assigned
Delivered	Received
Cancelled	Before dispatch

d. Commission on Mechanic Orders:

If mechanic orders for job:

Option A:

- Commission charged to vendor only

Option B:

- Special mechanic discounted rate (Configurable)

Admin configurable rule required.

4.2.1.9. Purchase History (Mechanic):

Mechanic can view:

- Past ordered products
- Frequently ordered parts
- Repeat order option
- Invoice download

Note- Only online payment will be allowed for the spare part.

4.2.1.10. Wallet Module:

Wallet can be used for:

- Receiving referral rewards
- Refunds

Mechanic can:



- View wallet balance
- Transaction history

4.2.1.11. Reports & Analytics (Mechanic):

- Daily earnings
- Weekly earnings
- Monthly summary
- Total completed jobs
- Cancellation ratio
- COD success rate

4.2.1.12. Notifications:

Mechanic receives push notifications for:

- New booking
- Order update
- Payment received
- Admin message
- Low rating alert

4.2.1.12. Profile:

- Profile
- Edit Profile
- Logout

4.3 .(Delivery Partner) Functional Requirements

4.3.1 Registration Fields:

- Full Name (Mandatory)
- Mobile Number (OTP Verification)
- Email (Optional)
- Profile Photo
- Aadhaar Number
- Driving License
- Vehicle Type (Bike / Cycle)
- Vehicle Number
- Bank Details (Mandatory)
- Service Area Radius

4.3.2 Approval Flow:

- Delivery partner submits registration
- Status = Pending
- Admin reviews documents
- Admin:
 - Approve
 - Reject (with reason)

Statuses:

- Pending
- Approved
- Rejected
- Suspended



4.3.2 Login:

Delivery Partner login via:

- Registered Mobile Number
- OTP Verification

4.3.3 Dashboard (Home Screen):

Features:

- Online / Offline Toggle
- Active Deliveries
- Today's Earnings
- COD Pending Amount
- Delivery History
- Notifications

4.3.3 Order Assignment:

a. Assignment Logic

Orders are assigned based on:

- Nearest Delivery Partner
- Availability Status
- Service Radius
- Current Active Deliveries

Assignment Type:

Type	Description
Automatic	System assigns nearest partner
Admin Override	Admin manually assigns

b. Order Notification Screen:

Delivery partner can view:

- Order ID
- Pickup Location (Vendor)
- Drop Location (User / Mechanic)
- Distance
- Estimated Delivery Time
- COD / Online
- Earning Estimate

Delivery partner must:

- Accept
- Reject

within a configurable time limit.

4.3.4 Delivery Flow:

a. Status Flow

Status	Updated By
Assigned	System
Accepted	Delivery Partner
Reached Pickup	Delivery Partner
Picked Up	Delivery Partner
Enroute	Delivery Partner
Reached Drop	Delivery Partner
Delivered	Delivery Partner
Cancelled	Admin

b. Delivery Steps:

- Accept order
- Navigate to vendor
- Mark "Reached Pickup"
- Collect parcel
- Mark "Picked Up"
- Start delivery
- Navigate to drop location
- Mark "Reached Drop"
- Complete delivery

4.3.7 Earnings & Settlement:

Delivery partner can view:

- Daily Earnings
- Weekly Earnings
- Monthly Earnings
- Completed Deliveries
- COD Pending
- Settlement Status

Admin settles payout periodically.

4.3.8 Notifications:

Push notifications for:

- New Order
- Order Cancelled
- Delivery Completed



- Admin Message

4.3.9 Delivery History:

Partner can view:

- Past Deliveries
- Order ID
- Pickup Location
- Drop Location
- Earnings
- COD Status
- Delivery Date

4.3.10. Ratings & Feedback:

After delivery:

- User / Mechanic rates partner
- Rating impacts assignment priority
- Partner can view:
 - Average Rating
 - Feedback comments

4.3.11. Wallet Module:

Wallet used for:

Partner can:

- View wallet balance
- Transaction history

4.3.12. Profile Management:

a. View Profile:

Delivery Partner can view the following profile details:

Field	Editable
Profile Photo	Yes
Full Name	Yes
Mobile Number	No
Email ID	Yes
Aadhaar Number	No
Driving License	No
Vehicle Type	Yes
Vehicle Number	Yes
Bank Details	Yes
Service Radius	Yes
Account Status	No
Rating	No

b. Edit Profile:

Delivery partner can update:

- Profile Photo
- Email ID
- Vehicle Type
- Vehicle Number
- Bank Account Details



- Service Radius

c. Logout



4.4 .(Vendor Panel) Functional Requirements

4.4.1 Vendor Registration & Onboarding:

a. Registration Fields:

- Business Name (Mandatory)
- Owner Name
- Mobile Number (OTP Verification)
- Email ID
- GST Number
- PAN Number
- Business Address
- Pickup Address
- Service Radius
- Bank Account Details

b. Approval Flow:

- Vendor submits registration
- Status → Pending
- Admin verifies documents

Admin:

- ❖ Approve
- ❖ Reject (with reason)



c. Login:

Vendor can login using:

- Registered Mobile Number
- Password

OR

- Registered Email ID
- Password

4.4.2 Vendor Dashboard:

Features:

- Total Orders
- Pending Orders
- Completed Orders
- Total Earnings
- Commission Deducted
- Active Products
- Low Stock Alert
- AI Pricing Suggestion Alerts

4.4.3 . Product Management Module:

a. Add Product:

Fields:

- Product Name
- Product Category
- Compatible Vehicle Type
- Brand
- Model Compatibility
- Description
- Product Images
- Base Price
- Discount
- Stock Quantity
- SKU
- Variant (Size / Model / Engine Type)

b. Product Variant Management:

Vendor can add:

Example:

Product	Variant	Stock
Engine Oil	1L	10
Engine Oil	3L	5

Stock auto reduces based on order.

c. AI Pricing Recommendation:

Vendor will see:

- Recommended Price
- Market Average Price
- Suggested Margin %
- Undercut Alert
- Overpricing Alert

Vendor can:

- ❖ Accept
- ❖ Modify
- ❖ Ignore



4.4.4 . Inventory Management:

Vendor can:

- View current stock
- Update stock
- Receive:
 - Low stock alert
 - Overstock alert
 - High demand alert

AI predicts:

- Fast moving products
- Dead stock risk
- Restock quantity



4.4.5 . Order Management:

Vendor receives orders from:

- User
- Mechanic

a. Order Status Flow:

Status	Updated By
Order Placed	System
Accepted	Vendor
Rejected	Vendor
Packed	Vendor
Ready for Pickup	Vendor
Cancelled	Admin

b. Order Processing Flow:

- New order received
- Vendor accepts order
- Packs product
- Marks "Ready for Pickup"
- Delivery partner auto assigned

Note - The countdown will start as per the order.

4.4.6 .Cancellation:

Vendor can:

- Cancel before packing
- Must provide cancellation reason

Refund handled by admin.

4.4.7 .Delivery Management:

Vendor can:

- View assigned delivery partner
- Track live delivery
- Contact delivery partner (In-App)

4.4.8 .Commission Management:

Product Commission:

- 5% – 10%
- Configurable by Admin

Deducted automatically from vendor payout.

Vendor can view:

- Gross Amount
- Commission



- Net Payout

Note - The commission will be defined by the admin.

4.4.9 .Earnings & Settlement:

Vendor can view:

- Daily Earnings
- Weekly Earnings
- Monthly Earnings
- Completed Orders
- Pending Settlement
- Settled Amount

4.4.10 .Reports:

Vendor can view/download:

- Sales Report
- Product-wise Revenue
- Order Report
- Settlement Report

4.4.11 .Notifications:

Vendor receives alerts for:

- New Order
- Order Cancelled
- Low Stock
- AI Pricing Suggestion
- Settlement Done



- Admin Message

4.4.12 .Profile Management:

Vendor can update:

- Business Name
- Address
- Pickup Location
- Bank Details
- Service Radius

Bank update requires OTP verification.

4.4.13 . Logout Module:

Vendor can logout from:

- Profile
- Dashboard

4.5 .(Admin Panel) Functional Requirements

4.5.1. Admin Login & Authentication:

a. Login Fields:

- Email ID
- Password
- OTP Verification (Optional)

b. Security Features:

- Role-Based Access Control
- Session Timeout
- Multi-device Login Control
- Login Activity Logs
- Password Reset

4.5.2. Dashboard:

Admin Dashboard shows:

- Total Users
- Active Mechanics
- Active Vendors
- Active Delivery Partners
- Total Bookings
- Product Orders
- Pending Refunds
- COD Pending Amount
- Total Commission Earned



- Total Revenue

Filters:

- Daily
- Weekly
- Monthly
- Custom Date Range

4.5.3. User Management:

Admin can:

- View Users
- Search User
- Block / Unblock
- Edit User Profile
- View Bookings
- View Product Orders
- Adjust Wallet Balance

4.5.4. Mechanic Management:

Admin can:

- Approve / Reject Registration
- View Documents
- Suspend Account
- Adjust Commission
- View Ratings
- Track Active Job

Mechanic Status:

- Pending
- Approved
- Rejected
- Suspended

4.5.5. Vendor Management:

Admin can:

- Approve / Reject Vendor
- Verify GST
- View Product List
- Suspend Vendor
- Adjust Commission %
- Freeze Settlement
- Modify Service Radius

4.5.6. Delivery Partner Management:

Admin can:

- Approve / Reject Delivery Partner
- View Documents
- Suspend Account
- Modify Service Radius
- Reassign Delivery
- Adjust Commission
- Track Active Delivery

4.5.7. Booking Management (Mechanic Service):

Admin can:

- View all bookings
- Cancel booking
- Reassign mechanic
- Modify service charges
- Track live mechanic

Booking Status:

- Pending
- Assigned
- Enroute
- Reached
- Work Started
- Completed
- Cancelled

4.5.8. Product Order Management:

Admin can:

- View orders
- Cancel order
- Approve return
- Approve refund
- Reassign delivery partner
- Modify delivery charge

Order Status:

- Placed
- Accepted
- Packed
- Out for Delivery
- Delivered
- Cancelled

4.5.9. Commission Management:

Admin can configure:

Module	Commission
Mechanic Service	12% or ₹150
Product Order	5-10%
Delivery	12% or ₹150

Commission applicable on:

- Mechanic
- Vendor
- Delivery Partner

4.5.10. Refund Management:

Admin can:

- Approve / Reject Refund
- Verify return request
- Initiate refund



Refund Mode:

- Online → Source Account
- COD → Wallet

4.5.11. Wallet Management:

Admin can:

- Credit wallet
- Debit wallet
- Provide referral reward

4.5.12. Referral Management:

Admin can:

- Configure reward amount
- Track referral usage
- View reward history

4.5.13. Delivery Assignment Control:

Assignment Types:

- Auto (Nearest Partner)
- Manual Override

Admin can:

- Change assigned partner

- Modify delivery route

4.5.14. Sub Admin

Admin can:

- Sub admin name
- Email
- Phone number
- Password
- Confirm Password
- Action
 - Continue Button
 - Add
 - Edit
 - Delete
 - Active/INactive

Note - The Sub admin will receive the password via a text message or the associated email address.

4.5.15. AI Configuration Panel:

Admin can:

- Set price recommendation sensitivity
- Set mechanic recommendation weight
- Configure inventory forecast
- Configure competitor analysis range

4.5.16. Reports & Analytics:

Admin can generate:

- Booking Report
- Order Report
- Commission Report
- Vendor Sales Report
- Mechanic Earnings Report
- Delivery Earnings Report
- Settlement Report

Export Options:

- PDF
- Excel

4.5.17. Notifications:

Admin can send:

- Push Notification
- Email
- SMS

To:

- Users
- Mechanics
- Vendors
- Delivery Partners

4.5.18. Profile Management:



Admin can update:

- Name
- Email
- Password

4.5.19. Logout Module:

Admin can logout:

System Behaviour:

- Session terminated
- Login required again



4.6 .(AI Module) Functional Requirements

4.6.1. AI-Based Dynamic Product Pricing (Vendor Panel):

AI Engine will analyze:

- Product price in a particular area
- Vendor location
- Historical sales
- Competitor pricing
- Product availability
- Delivery timeline
- Service area demand

And recommend an **optimal selling price** for auto spare parts to vendors.

Functional Flow:

- Vendor uploads product
- Vendor enters base price
- AI Engine analyzes:
 - Nearby vendor price
 - Market demand
 - Stock availability
 - Past purchase trends
- AI suggests:
 - Recommended Selling Price
 - Minimum Competitive Price
 - Maximum Margin Price
- Vendor can:
 - Accept suggested price
 - Edit manually
 - Reject suggestion

AI Output Fields:

Field	Description
Recommended Price	AI suggested price
Market Avg Price	Avg nearby vendor price
Suggested Margin %	Profit suggestion
Demand Index	High / Medium / Low
Competitive Score	Market comparison score

4.6.2 .AI-Based Inventory & Unit Management:

AI will predict:

- Product demand
- Fast-moving parts
- Low-stock alerts
- Dead stock risk

Functional Flow:

- AI monitors:
 - Order frequency
 - Vendor stock
 - Product movement
- AI predicts:
 - Future demand
 - Required stock
- Vendor notified:
 - Restock Alert



- Overstock Alert
- Suggested Order Quantity

Inventory AI Alerts:

- Low Inventory Warning
- High Demand Prediction
- Slow Moving Product
- Out of Stock Forecast

4.6.3 .AI-Based Mechanic Recommendation Engine (User App):

AI will recommend the **best mechanic** based on:

- Overall Rating
- Service Quality Score
- Completion Rate
- Response Time
- Distance
- Service Type Expertise
- Cancellation Ratio

Functional Flow:

- User creates booking request
- AI Engine ranks nearby mechanics
- Score calculated using:

(Mechanic Score = (Rating * 30%) +
(Service Completion * 20%) +(Response Time * 15%) +
(Distance * 15%) +(Expertise Match * 10%) +(Cancellation Rate * 10%)

- Top 3 Recommended Mechanics shown

4.6.4 .AI-Based Vendor Competitive Pricing Analysis:

AI continuously monitors:

- Vendor Product Pricing
- Product Demand
- Regional Price Trends
- Purchase Behavior

And informs vendors about:

- Competitor Price Range
- Market Price Gap
- Suggested Competitive Price
- Undercut Alert
- Overpricing Alert

Vendor Dashboard Insights:

- You are ₹120 higher than nearby vendors
- Reduce price by 5% to improve order probability
- Market demand is increasing for this product
- Suggested new price: ₹2,450